



Truncating

Task A: Truncating

1) Complete the table.

Number	Truncate to 1 significant figures	Truncate to 2 significant figures	Truncate to 1 decimal place
89 231.625	80 000	89 000	89 231.6
0.05681	0.05	0.056	0.0
0.0002563	0.0002	0.00025	0.0
23.145	20	23	23.1
19.99999	10	19	19.9

2) Fill in the cross number puzzle.

¹ 5				⁶ 1
² 6	3	⁴ 2	0	0
0		3		.
³ 0	.	3	2	⁵ 1
0		0		4

Across

2) **Truncate** 63215 to 3 s.f.

3) **Truncate** 0.321986 to 3 d.p.

Down

1) **Truncate** 56842 to thousands.

4) **Truncate** 2331.124 to 3 s.f.

5) Round 13.856 to 2 s.f.

6) Round 10.13986 to 2 d.p.

3) Work out the number given the clues.

a) I am

252

Truncated to 1 s.f. I am 200

I am a multiple of 12

I am 250 when rounded to the nearest 25

I am a palindrome (number reads the same if all the digits reversed)

b) I am

168

Truncated to 1 s.f. I am 100

I am a multiple of 4

I am 200 when rounded to the nearest 100

The sum of my digits is 15

Task B: Truncating in real life

1) Given the scenarios below, identify if the answer should be given exactly, rounded or truncated.

Aisha is 14.7 years old and wants to see a 15 Rated film. Can she go?

No, the truncated age is 14 years old so she is below age.

A book shop has £89 and wants to buy as many £1.99 puzzle books as possible. How many books can it buy?

Truncate because it can only afford integer number of books.

A bill of £89 is shared between 10 people.

Round because money is 2 d.p. This means each person should pay £8.90

(Or pay £9 each and leave a tiny tip!)

2) Here is a broken calculator screen.



What is the largest and smallest number that could be displayed on the screen?

Given the number is concealed, we see the truncated value.

The largest value is 1.2999999999... and the smallest is 1.2



3) In computer science, 'INT' or 'int' is used to **truncate** a number so to only give the integers.

For example,

$\text{INT}(56/10)$ will have an output of 5

Work out the output of the following.

- | | |
|--|---|
| a) $\text{INT}(45/8)$ | <i>Output is 5</i> |
| b) $\text{INT}(98/10) + \text{INT}(7.8/3)$ | <i>Output is 11 because $9 + 2$</i> |
| c) $\text{INT}(8/0.5) + \text{INT}(12.65/2) - \text{INT}(9.8/6)$ | <i>Output is 21 because $16 + 6 - 1$</i> |

4) Lucas is going on holiday and looks at the exchange rates.

$\text{£}1 = 1.16887$ Euros

Travel firms and banks offer a truncated value to this:

$\text{£}1 = 1.16$ Euros

a) Why is truncating exchange rates better for travel firms and banks?

For every £1 exchanged the travel company or bank gain 0.00887 euros

b) If £1 000 000 was exchanged, how much money would the travel companies make due to truncation?

Multiply 0.00887 by 1 000 000 gives €8870 gain, so for every £1 000 000 exchanged they earn €8870 due to truncation.

